

3 Data-driven Computed Tomography Reconstruction

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Key Publication	See list below

Project Description

Computed Tomography (CT) is a widely used imaging technique in medicine and industry, but the process of reconstructing CT images is complex. CT scanners acquire multiple X-ray projections (similar to X-ray images you see when you break a bone) from different angles, but these projections alone don't reveal depth information. To recover that, the tomography machine rotates and acquires several projections. From this set of images (a.k.a. *sinograms*) obtaining the CT reconstructed image is a mathematically very challenging problem, as the problem becomes *ill-posed*, when sinograms are sparsely sampled. As a result, a large number of X-ray projections is required to avoid reconstruction errors, leading to prolonged exposure times. This higher radiation dose is undesirable, especially in medical applications.

Nowadays, the literature on using machine learning to solve ill-posed problems like CT is booming. Most recent methods claim to require a reduced X-ray dosage while producing the same image quality. These methods range from those grounded in mathematical theory to those focused on computationally scalable approaches.

In this project, you will select and reproduce a key paper from the literature on using machine learning for low-dose CT reconstruction and to add it to the LION toolbox (github.com/CambridgeCIA/LION).

Project goals

Main goals of the project:

1. Produce a pipeline for loading the [Mayo Clinic Low Dose CT Grand Challenge dataset](#) (or similar datasets), producing simulated low-dose measurements, and reconstruct it using traditional non-ML algorithms, using [LION](#)
2. Implement one learned reconstruction method from the list below (or propose a new one).
3. Reproduce the experiments, results and plots from the chosen paper.

Method	Math	Code	Physics	ML
Patch-based diffusion models	70%	40%	80%	80%
Regularising Inverse Problems with Generative Machine Learning Models	90%	50%	50%	50%
RISING	40%	40%	10%	50%
Neural Field Regularization by Denoising	50%	30%	30%	50%
Stochastic Learned Primal Dual	40%	50%	10%	50%

Table 1: Difficulty estimate for each possible method.

The above goals will get you a high grade in the project. Note that the projects available are of different difficulty level (see Table 1) and we will take that into consideration at grading. More

options are available, and we welcome proposals from the students. Possibility of enhancing the models and publication of results exist, depending on how far you get in implementation and development.

These should be optional extensions for stronger students to attempt if they manage to complete the main project goals. If students complete **one** of these to a high standard then they should be typically awarded a high distinction. Those could be things like updating or changing the methods, adding a new features or functionality to the analysis pipeline, or including new data to the analysis.

Up to 5 students can be selected for this project.

Prerequisites

You need to take the Medical Imaging minor, or have equivalent pre-existing knowledge. Strong coding skills (`pytorch`, `scipy`, `numpy`) are required, but much of this will be taught in the MPhil. For the more “mathy” sub-projects, a strong maths background, particularly in optimization, is desired.

Reading List

A general book on CT that is good to have in hand:

- Hansen, Per Christian, Jakob Jørgensen, and William RB Lionheart, eds. **Computed tomography: algorithms, insight, and just enough theory.** *Society for Industrial and Applied Mathematics*, 2021.

And of course, the corresponding papers linked for each project. More references and reading list will be given once we start the project.

Hope you join us!